



R o b o t i q u e
CRC
R o b o t i c s

CRC
**PROGRAMMING
COMPETITION**

MODUEL

BLOC C

A FEW NOTES

- The complete rules are in section 4 of the rulebook.
- You have until **18h29** to submit your code.
- **We are giving you quick and easy-to-use template files for your code and the tests. You are required to use them.**

USING THE TEMPLATE FILE

- The template tests call the function to test, take the output and allow you to quickly check if your code works as intended. **All your code, except additional functions you create, should be written in the function of the part of the problem you are solving.**
- Points given in the document are indications of how difficult the section is and how many points you will get if you complete it. This preliminary problem is going to be the same amount of points as the total of Bloc A without any multiplier. (500 points)
- There is no multiplier for Bloc C

STRUCTURE

Every problem contains a small introduction like this about the basics of the problem and what is required to solve it.

Input and output specification:

Contains the inputs and their format, and which outputs the code is required to produce and in what format they shall be.

Sample input and output:

Contains a sample input, sometimes containing sub examples in the sample input, and what your program should return as an output.

Explanation of the first output:

Explains briefly the logic that was used to reach the first output, given the first input.

Crab escape!

In this special problem from Block C, you must control crabs so that the designated crab reaches the exit with the least total movement of all crabs combined.

You are given a 6×6 grid on which all the crabs are placed. On this grid, crabs can only move sideways relative to their orientation. A crab positioned horizontally on the grid can only move left and right. Similarly, a crab positioned vertically can only move up and down. The objective is to allow Crab 1 to exit the grid by reaching the exit square.

The crabs will be provided for each example, but the exit position will always remain the same. You must always get Crab 1 out by moving it so that it reaches the exit square at position (5, 2). Crab 1 will always have dimensions 2×1 and will, of course, start on the same row as the exit. The grid squares range from 0 to 5 on the x-axis, with 0 being the leftmost column and 5 being the rightmost column. On the y-axis, square 0 is the top row and square 5 is the bottom row.

You must find the solution with the least total movement and you must do so as quickly as possible!!

Input and output specifications:

Input specifications:

You will receive as input a grid of integers, where 0 represents an empty cell, 1 represents Crab 1, 2 represents Crab 2, and so on.

Crabs may have dimensions 1×1 , 2×1 (or 1×2), or 3×1 (or 1×3). Crabs of size 1×1 may move either horizontally or vertically, as long as they do not leave the grid or collide with another crab. Crabs of size 2×1 and 3×1 move only sideways relative to their orientation and always keep that same orientation.

Output specifications:

As output, you must provide a series of commands in the form of a list of strings. Each command consists of two elements. The first element is the number of the crab to be moved, and the second element is the letter indicating the direction in which it should slide (up = U, down = D, left = L, and right = R).

For example, if Crab 1 must move to the right, the command would be "1R". Crab 1 will always move horizontally (from left to right), so a command such as "1U" is invalid, since Crab 1 has dimensions 2×1 .

Scoring (500 points)

The scoring is separated into 2 parts, which are the precision score (amount of moves used to solve the challenge) and the speed score (time of execution). For each challenge configuration, a maximum of 50 points will be awarded for the precision score and 20 points for the speed score, for a total of 70 points per configuration. If your program does not return a valid solution within the **10 seconds of execution allowed for an example**, the score for that example will be 0. **It is strictly forbidden to use multithreading and your code must fully run on CPU.** Your program will be tested on the same 10 grids for all schools to give out your final score.

Precision Score

The precision score is the total amount of moves that your crabs have to make in order to solve the challenge grid. If you have the solution with the least amount of steps out of all the schools, you will receive 50 points. Every additional move over the best solution will lose you 5 points until a score of 0 if you have more than 10 moves over the best solution.

Speed Score

The speed score measures the speed of your program compared to the other schools. You have a maximum of 20 points if you find a valid solution the fastest. You lose 1 point per ranking, so only the first 20 teams get points for the speed score. To be considered in the speed ranking for a configuration, your solution has to be valid.

Number of moves (lowest amount = X)	Precision score
X + 0 moves	50
X + 1 moves	45
X + 2 moves	40
...	...
X + 9 moves	5
X + 10 moves	0
> X + 10 moves	0

Speed compared to other schools	Speed score
fastest	20
second fastest	19
third fastest	18
...	...
19th fastest	2
20th fastest	1
21st and more	0